Computational Paradigms in 4D Design Software: Spatio-Temporal Process Simulation in AECO and Inverse Material Programming in Additive Manufacturing

Author: SAMUELSON G

Role: Principal Investigator and Independent Researcher and Sole Author

Correspondence: Correspondence and requests for materials, computational schemas, or comparative benchmark matrices should be addressed to SAMUELSON G.

Email: gsamuelsonguna@gmail.com

ORCID: 0009-0005-8744-8178

Abstract

Four-dimensional (4D) design software encompasses computational environments that integrate temporal dynamics (t) directly into three-dimensional (x, y, z) geometric and semantic representations. Across contemporary engineering and materials science, 4D software bifurcates into two distinct yet mathematically analogous domains: macro-scale spatial-temporal construction simulation via 4D Building Information Modeling (4D BIM) in the Architecture, Engineering, Construction, and Operations (AECO) sector, and micro-to-meso-scale chemo-mechanical morphogenesis via 4D printing (4DP) inverse design software in advanced additive manufacturing. This study presents an exhaustive technical synthesis and empirical evaluation of both paradigms. In the AECO domain, the analysis examines OpenBIM Industry Foundation Classes (IFC) temporal schemas (IfcWorkSchedule, IfcTask, IfcRelAssignsToProcess), spatial-temporal workspace clash detection algorithms, and constraint-driven generative scheduling engines. Empirical benchmarks from multi-storey reinforced concrete, modular prefabrication, and subterranean pipeline projects demonstrate that integrated 3D+4D+5D BIM workflows elevate baseline composite accuracy from 78.4% (manual methods) to 96.7%, compress post-change re-estimation turnaround times by 97.1% (from 42.0 hours to 1.2 hours), reduce modular site durations by 50% (from 18 days to 9 days), and eliminate up to 100% of spatial-temporal pipeline collisions. In the additive manufacturing domain, the analysis investigates the transition from forward kinematic self-assembly simulators (such as Project Cyborg) to deep learning inverse design architectures—specifically Fully Convolutional Networks (FCNs), normalizing flows (4DYNAMO), and semantic segmentation networks—that map target 3D deformed

morphologies directly into fabrication-ready voxel material layouts and liquid crystal elastomer (LCE) bilayer orientations in a single non-iterative inference pass. Finally, emerging intellectual property frameworks governing AI-generated 4D transformations and G-code manufacturing instructions are evaluated under contemporary patent jurisprudence.

Keywords: 4D Design Software; 4D Building Information Modeling (4D BIM); Spatial-Temporal Clash Detection; Generative Construction Scheduling; 4D Printing (4DP); Non-Iterative Inverse Design; Industry Foundation Classes (IFC); Programmable Matter; Cyber-Physical Digital Twins; Invertible Normalizing Flows.

Taxonomic Foundations of Four-Dimensional Design Computation

Traditional Computer-Aided Design (CAD) and static 3D modeling treat physical artifacts and built facilities as time-invariant Euclidean geometries. The introduction of the fourth dimension—time—fundamentally alters the computational representation of an object from a static boundary representation (B-Rep) or voxel mesh into a time-parameterized state function $\mathcal{M}(\mathbf{x}, t)$, where $\mathbf{x} \in R^3$ denotes spatial coordinates and $t \in [t_0, t_f]$ denotes a discrete or continuous temporal domain. Despite operating at vastly different physical scales, 4D design software in construction engineering and 4D design software in smart additive manufacturing solve an identical fundamental problem: predicting, optimizing, and controlling how physical matter occupies, traverses, and transforms within three-dimensional space over time.

In the macro-scale AECO paradigm, 4D BIM software models discrete kinematic assembly and workspace occupancy. Individual structural, architectural, and mechanical-electrical-plumbing (MEP) elements remain geometrically rigid, but their existence state (unbuilt, under construction, temporary work, completed, or demolished), spatial trajectory (crane hoisting, modular placement), and surrounding operational buffer zones evolve dynamically according to a project schedule network. Conversely, in the meso- and micro-scale additive manufacturing paradigm, 4D printing design software models continuous chemo-mechanical morphogenesis. Originating with the foundational self-assembly research introduced by Skylar Tibbits at the MIT Self-Assembly Lab in collaboration with Stratasys and Autodesk between 2013 and 2014, 4D printing embeds a transformation program directly into multi-material 3D-printed structures. When exposed to external environmental stimuli—such as thermal fields, moisture or water immersion, pH gradients, light, or electromagnetic radiation—differential internal strains induce programmable folding, curling, swelling, or stiffness switching without external electromechanical actuators.

Table 1: Taxonomic Comparison of Macro-Scale 4D BIM and Meso-Scale 4D Printing Software Architectures

Computational Dimension	Macro-Scale AECO: 4D BIM & Generative Scheduling	Meso/Micro-Scale AM: 4D Printing & Inverse Design CAD
Primary Physical Scale	Meters to kilometers (10^0m to 10^4m); buildings, infrastructure, underground pipelines.	Micrometers to meters (10^{-6} m to 10^0m); biomedical stents, metamaterials, soft robotics, gridshells.
Temporal Transformation Mechanism	Discrete sequential assembly, resource allocation, workspace progression, and equipment kinematics.	Continuous non-linear deformation, anisotropic swelling, shape-memory recovery, and self-assembly.
Governing Mathematical Formulation	Directed acyclic graphs (CPM), Line-of-Balance (LOB), spatial-temporal bounding-box intersection, constraint programming.	High-dimensional non-linear continuum mechanics, diffeomorphic normalizing flows, voxel semantic segmentation.
Core Data Exchange Standards	Industry Foundation Classes (IFC), BIM Collaboration Format (BCF), Native RVT/NWD/DGN, Primavera XML/XER.	Multi-material voxel matrices, slice label maps, orientation tensors, additive manufacturing G-code.
Representative Software Ecosystem	Bentley SYNCHRO 4D Pro, Autodesk Navisworks Manage, BEXEL Manager, Kalloc Fuzor, ALICE Technologies.	Autodesk Project Cyborg, 4DYNAMO, 4DMesh, FCN/Segmentation Inverse Frameworks, FEA-CAD pipelines.

Computational Dimension	Macro-Scale AECO: 4D BIM & Generative Scheduling	Meso/Micro-Scale AM: 4D Printing & Inverse Design CAD
Primary Optimization Objective	Minimize project duration, labor/equipment cost, and spatial-temporal trade collisions ($CI \rightarrow 0$).	Minimize Mean Relative Deformation Error (MRDE) between actuated material layout and prescribed target 3D geometry.

OpenBIM Data Schemas and Algorithmic Mechanics in 4D Construction Software

Industry Foundation Classes and Temporal Process Ontologies

The operational efficacy of 4D BIM software depends on seamless semantic interoperability between three-dimensional parametric authoring platforms and temporal project management engines. Within traditional fragmented delivery frameworks, architectural and structural models are authored independently from Critical Path Method (CPM) schedules, resulting in static Gantt charts that rapidly desynchronize from site realities. To establish a vendor-neutral Single Source of Truth within a Common Data Environment (CDE), the buildingSMART Industry Foundation Classes (IFC) schema formalizes an explicit ontology for 4D process and control integration.

Within the IFC data model, temporal planning is governed by the IfcControl and IfcProcess class hierarchies. At the macro-management level, the IfcWorkPlan entity encapsulates top-level project planning metadata, while IfcWorkSchedule acts as a specialized sub-container that organizes construction activities into a structured timeline. Individual schedule activities are instantiated as IfcTask entities (a subtype of IfcProcess), which store temporal attributes including task duration, early and late start and finish dates, applied calendars, constraints, and total float. The structural linkage between management control and process execution is established via the objectified relationship entity IfcRelAssignsToControl, which binds IfcTask instances to their governing IfcWorkSchedule.

Crucially, 4D visualization and simulation require bidirectional relational pointers between schedule tasks and physical 3D building components. This mapping is executed through IfcRelAssignsToProcess, an objectified relationship that connects IfcTask entities directly to IfcBuildingElement instances (representing physical products such as slabs, columns, or curtain walls) and IfcResource entities (representing

labor crews, construction equipment, and consumable materials). Advanced openBIM software suites leverage Information Delivery Specifications (IDS) and automated data enrichment engines to verify these relational links before simulation, ensuring that large monolithic BIM objects—such as continuous multi-zone floor slabs—are algorithmically split according to custom construction zone rules to match the granularity of leaf-level IfcTask work packages.

Spatial-Temporal Clash Detection and Workspace Congestion Formulations

While 3D BIM coordination identifies static geometric intersections—commonly classified as hard clashes (direct physical overlap between solid elements, such as a structural steel beam penetrating an HVAC duct) and soft or clearance clashes (violations of minimum maintenance, insulation, or installation buffer tolerances)—it remains blind to time-dependent conflicts. 4D BIM software introduces workflow, operational, and spatial-temporal clash detection, joining spatial verification with sequence validation so planners can detect not only whether physical components collide, but whether construction activities are scheduled in an unbuildable sequence or compete for the same three-dimensional workspace simultaneously.

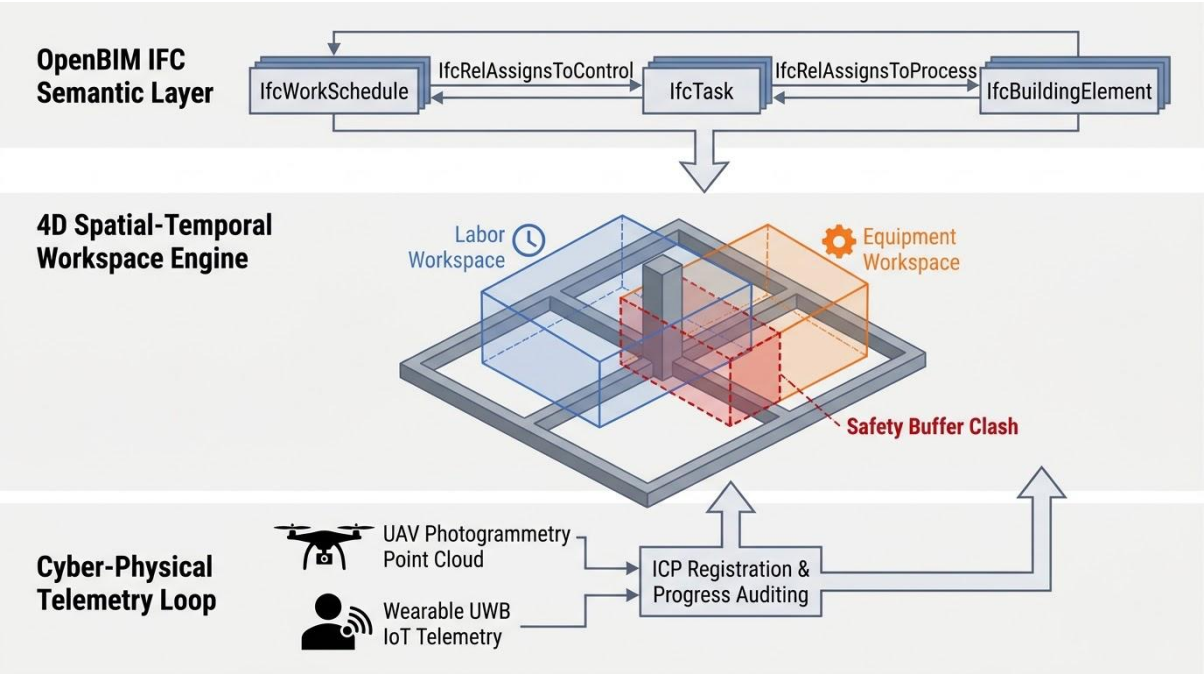


Figure 1: End-to-End Computational Architecture of 4D BIM and Cyber-Physical Site Synchronization

The architecture integrates three operational layers: (Top) OpenBIM semantic federation linking physical elements to temporal tasks via Industry Foundation Classes; (Middle) Dynamic 4D workspace collision analysis distinguishing static hard clashes from temporal labor and equipment buffer overlaps; and (Bottom) Cyber-physical

synchronization matching as-built point clouds and worker coordinates against planned 4D milestones.

To execute spatial-temporal interference checking mathematically, 4D software engines discretize both physical building elements and transient construction workspaces into volumetric primitives. For static and dynamic interference analysis, the Bounding Box Method encapsulates an element within an axis-aligned or oriented bounding prism, whereas the Component Decomposition Method subdivides complex geometries into smaller constituent sub-elements, computing the individual bounding box volume of each sub-component and summing them to capture true occupied space rather than mere material quantity takeoff (QTO) volume. To calculate total floor or zone availability (V_{zone}), 4D engines apply a Boolean mesh discretization protocol wherein the overarching floor bounding box is partitioned into uniform grid cells of side length Δs, cells intersecting structural footprints are marked as covered, and the total occupied area is multiplied by the floor height H_f.

During 4D simulation, daily space snapshots generate transient 3D workspaces linked to active schedule tasks across four functional categories: labor workspaces, equipment workspaces, material storage workspaces, and physical distance (safety buffer) workspaces. When pairwise intersection tests run across daily time steps, naive geometric checking generates severe false-positive noise—such as flagging unrealistic clashes when a non-physical safety distance buffer extends beyond a room boundary and intersects a static concrete wall. Advanced 4D engines resolve this by implementing semantic rule dictionaries that filter out redundant buffer-to-structure intersections and evaluate true operational congestion using the Space Loading Factor (SLF). For any construction zone z during time interval Δt, the SLF is defined as the ratio of cumulative required workspace volume for all concurrent activities $i \in \mathcal{A}_z(\Delta t)$ to the available zone volume:

$$\text{SLF}(z, \Delta t) = \frac{\sum_{i \in \mathcal{A}_z(\Delta t)} V_{\text{req},i}(t)}{V_{\text{avail},z}}$$

Zones exhibiting a high SLF experience severe spatial congestion, which amplifies the unitless Clash Impact metric by intensifying the probability of productivity degradation and occupational hazard transmission between overlapping crews.

Furthermore, because large-scale BIM models frequently generate thousands of raw geometric and temporal collisions, modern 4D coordination platforms incorporate automated prioritization and machine learning filtering algorithms. Expanding upon earlier four-parameter formulations, Bitaraf et al. established a six-parameter multiplicative Clash Criticality Index (CI) calibrated via the Best-Worst Method (BWM) to rank hard and spatial clashes during design and pre-construction:

$$CI = P \times W_S \times W_{\text{MEP(Time)}} \times W_{\text{MEP(Cost)}} \times N \times R \times U$$

In this formulation, P represents the geometric penetration depth between intersecting elements extracted from the clash engine, W_S denotes the structural element criticality weight (where beams and columns carry the highest structural weights), $W_{\text{MEP(Time)}}$ and $W_{\text{MEP(Cost)}}$ represent the schedule and financial consequence weights of the intersecting MEP system derived via BWM expert matrices, N captures the local spatial density of nearby hard and clearance clashes within a specified radius, R denotes the element-to-level volume ratio computed via the Boolean mesh algorithm, and U accounts for the functional criticality of the facility zone. Complementing mathematical ranking, hybrid filtering architectures that combine rule-based domain logic with supervised machine learning classifiers achieve a predictive accuracy of 0.96 in automatically isolating and suppressing irrelevant false-positive clashes—representing a 6% to 17% performance improvement over standalone or ensemble machine learning classifiers.

Comparative Evaluation of Commercial 4D BIM and Generative Scheduling Platforms

Architectural Divergence in Legacy Coordination versus Model-Based Scheduling

The commercial software ecosystem for 4D construction design and simulation has evolved from post-hoc visual playback tools into integrated, data-driven project control and generative optioneering environments. While all major platforms ingest 3D geometry and link elements to schedule tasks, comparative benchmarking reveals profound architectural differences in how each software handles native 3D manipulation, schedule authoring, automated rule mapping, and virtual reality (VR) site kinematics.

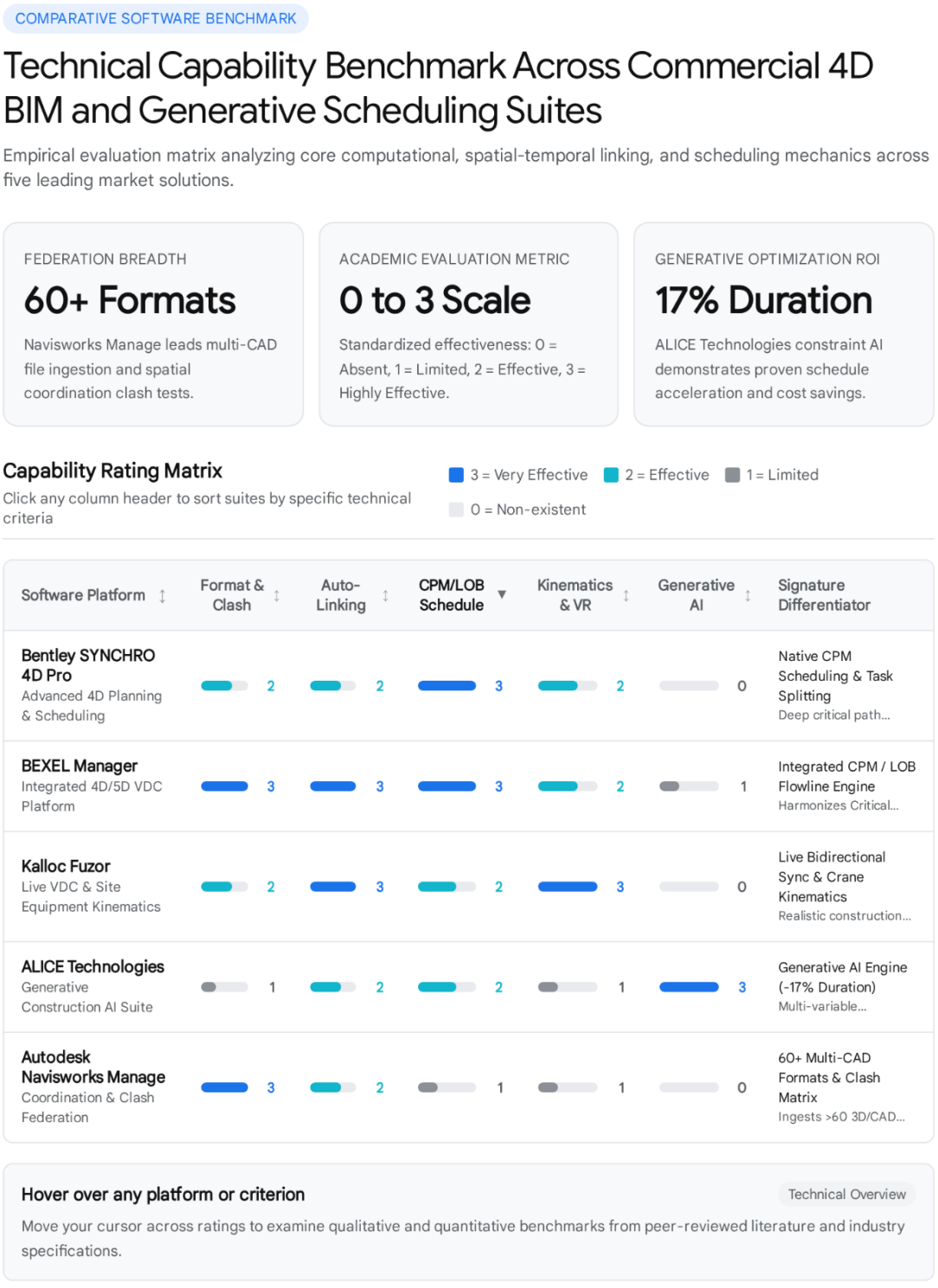


Figure 2: Technical Capability Benchmark Across Commercial 4D BIM and Generative Scheduling Suites

Evaluation scores reflect standardized 0–3 effectiveness benchmarks across core 4D workflows (0 = non-existent, 1 = limited, 2 = effective, 3 = highly effective). While Navisworks leads in broad file format ingestion (60+ formats), SYNCHRO 4D and BEXEL Manager excel in native CPM/LOB schedule authoring, Fuzor dominates

realistic site equipment kinematics, and ALICE Technologies pioneers constraint-based generative scenario optimization.

Data sources: BME OMIKK Academic Repository, The Future 3D Review, BEXEL Consulting, ALICE Technologies

Comparative empirical evaluations utilizing a standardized four-tier effectiveness rubric (0 = criterion does not exist, 1 = exists but is not very effective, 2 = exists and is effective, 3 = exists and is very effective) illuminate the distinct operational trade-offs among the primary commercial engines. Positioned primarily as a multi-disciplinary model federation, coordination, and clash review platform, Autodesk Navisworks Manage stands out for its exceptional format breadth, supporting native compatibility with over 60 distinct file formats including NWF, NWD, RVT, IFC, and DWG. Its TimeLiner module enables users to import external schedules from Microsoft Project, Primavera P6, or structured Excel CSV templates, or create basic calendar activities and link them to 3D selection sets. However, Navisworks lacks intrinsic capabilities to author or modify 3D model geometry or material definitions directly within the coordination environment, and its schedule engine functions primarily as a visual playback and clash-checking layer rather than a comprehensive CPM or resource-leveling environment.

In contrast, Bentley SYNCHRO 4D Pro is purpose-built from the ground up for model-based construction planning and 4D scheduling, integrating a full-featured CPM scheduling engine directly alongside the 3D viewport. Unlike coordination-first tools, SYNCHRO allows planners to split 3D geometry, manage complex task dependencies, allocate and level resources across labor, equipment, and materials, and integrate real-time Internet of Things (IoT) field sensor telemetry to update schedules dynamically based on actual site progress. Meanwhile, Kalloc Studios Fuzor—derived from a high-performance real-time gaming and physics engine—distinguishes itself through bidirectional live synchronization plugins and sophisticated site logistics animation. In Revit workflows, Fuzor's synchronization request plugin allows users to modify material types or element placements in Fuzor and selectively push listed modifications back to the native authoring model. Fuzor also enables rule-based automatic linking of 3D elements to schedule tasks using level filters, category filters, and custom parameters. In comparative testing involving tower cranes and concrete mixers, Fuzor provides superior kinematic realism, allowing planners to animate complex equipment trajectories, object manipulation, and material storage staging within an interactive Virtual Reality (VR) environment. Nevertheless, empirical tests note occasional shader translation anomalies—such as rendering glazed curtain wall panels as opaque blue surfaces upon initial import—and hierarchical selection tree complexities where clicking an element selects the entire model, necessitating manual navigation through the selection tree.

Representing an integrated openBIM 3D/4D/5D/6D enterprise platform, BEXEL Manager combines automated Information Delivery Specification (IDS) data verification, containment clash analysis, and an Intelligent 4D/5D Scheduling Engine. Rather than manually linking thousands of objects to tasks, planners define spatial Construction Zones (such as stories, phases, or structural grids) and reusable Construction Methodologies (logical templates defining trade sequence rules). The engine intersects these spatial and methodological matrices to automatically generate fully linked CPM and Line-of-Balance (LOB) or Flowline schedules, optimize Takt planning, split large model elements according to custom zone rules, and compute real-time 5D cashflow and monthly payment certificates. Within its Schedule Editor and Task Editor modules, planners manage working calendars, total float, resource leveling, and custom breakdown color-coding across active, completed, and critical-path elements. For smaller projects containing up to 15,000 BIM elements, BEXEL Manager Lite provides identical algorithmic functionality, while BEXEL TeamWorks and BEXEL CDE synchronize multi-user updates, document versioning, and BCF issue tracking across cloud dashboards and Power BI analytics.

Table 2: Feature and Algorithmic Matrix of Leading 4D BIM and Generative Scheduling Software Suites

Software Platform	Primary Architectural Paradigm	Schedule Authoring & Optimization Engine	3D-to-Task Linking Mechanism	Clash Detection & Workspace Analysis	Unique Technical Differentiator
Autodesk Navisworks Manage	Multi-disciplinary model federation & clash review.	Basic activity creation in TimeLiner; relies on external P6/MSP/Excel imports.	Manual selection/search sets and rule-based attribute matching.	Industry standard for 3D hard, soft/clearance, and basic 4D timeline clash tests.	Compatibility with over 60 native CAD/BIM file formats.
Bentley SYNCH	Dedicated model-	Native full CPM	Parametric auto-	Dynamic spatial-	Deep infrastruct

Software Platform	Primary Architectural Paradigm	Schedule Authoring & Optimization Engine	3D-to-Task Linking Mechanism	Clash Detection & Workspace Analysis	Unique Technical Differentiator
RO 4D Pro	based 4D construction planning & control.	engine, resource loading, and IoT-linked adaptive updating.	matching, user-field rules, and native 3D model splitting.	temporal conflict detection and real-time progress tracking.	ure scalability and live IoT digital twin synchronization.
Kalloc Studios Fuzor	Real-time VR physics simulation & site logistics kinematics.	Integrated 4D schedule editor with external schedule synchronization.	Rule-based auto-linking via level, category, and parameter filters.	Interactive hard, clearance, and 4D equipment path proximity analysis.	Bidirectional Revit change-sync plugin and realistic crane/mixer animations.
BEXEL Manager	Integrated openBIM 3D/4D/5D/CDE enterprise management.	Automated zone + methodology smart scheduling, CPM, LOB/Flowline, and Takt.	Automated smart mapping via custom breakdown structures and element splitting.	Custom clash matrices, containment clash API, and BCF server automation.	Simultaneous 4D/5D schedule generation and native Power BI / IDS verification.

Software Platform	Primary Architectural Paradigm	Schedule Authoring & Optimization Engine	3D-to-Task Linking Mechanism	Clash Detection & Workspace Analysis	Unique Technical Differentiator
ALICE Technologies	Explainable AI generative construction optioneering.	Constraint-based generative solver generating thousands of Pareto schedules.	Parameterized "recipes" attached to 3D BIM objects, block diagrams, or P6/MSP.	Proactive elimination of resource, crane radius, and precedence bottlenecks.	Automated "what-if" scenario simulation reducing project durations by ~17%.

Generative Construction Optioneering and Parametric Constraint Solving

A fundamental limitation of conventional 4D BIM tools (such as Navisworks and standard SYNCHRO workflows) is that they operate as descriptive or diagnostic validators: a human planner must first conceive a single execution logic, manually build the schedule network, and use the 4D software to visualize the sequence. Because a complex capital project contains millions of viable sequencing permutations—analogous to the 10^{40} viable board states in chess—human planners can manually test only a handful of "what-if" scenarios, leaving projects vulnerable to suboptimal resource allocation and fragile critical paths.

To overcome this cognitive bottleneck, Generative Construction Scheduling platforms—pioneered commercially by ALICE Technologies—decouple planning (governed by the human expert) from scheduling (executed autonomously by machine algorithms). Crucially, rather than relying on black-box deep neural networks—where 98% to 99% vision-recognition accuracy still leaves a 1% to 2% margin of unexplainable structural error, such as scheduling a roof slab before its supporting columns—generative construction engines utilize Explainable AI grounded in Constraint Programming (CP) and evolutionary genetic optimization. Users parameterize the construction process by assigning modular construction "recipes" (defining required operations, crew sizes, production rates, equipment requirements,

crane radii, calendars, and physical support precedence logic) to 3D BIM elements, 2D block diagrams, CSV files, or existing Primavera P6 and Microsoft Project schedules.

The constraint solver then explores the combinatorial state space to generate thousands of buildable, resource-loaded 4D schedules simultaneously, plotting the resulting solutions along a multi-objective Pareto front with project duration on the horizontal axis and total construction cost on the vertical axis. Project controls teams can instantaneously modify global parameters—such as adding a second tower crane, authorizing weekend overtime, or altering concrete curing times—and re-solve the entire schedule network in minutes to isolate the Pareto-optimal execution strategy. Across empirical capital project deployments spanning cast-in-place reinforced concrete structures, data centers, energy plants, and transit infrastructure, generative 4D scheduling has demonstrated an average 17% reduction in overall project duration, a 14% reduction in labor costs, and a 12% reduction in equipment costs. Furthermore, when paired with deep-learning schedule risk analytics platforms such as nPlan—which trains on over 300,000 historical project schedules to predict activity delay probabilities with up to 88% accuracy, while ALICE achieves approximately 85% efficacy in generative delay mitigation and recovery planning—4D design software transitions from a passive visualization aid into a closed-loop predictive control system.

Empirical Performance Benchmarks of 4D and 5D BIM Workflows

Design-Change Resilience and Estimation Turnaround Efficiency

To quantify the operational impact of integrating 4D temporal scheduling and 5D cost estimation into 3D parametric design models, comparative empirical evaluations have benchmarked four distinct project delivery methodologies across multi-storey reinforced concrete (RC) building construction: Manual Quantity Takeoff and Scheduling (M-QTO), 2D/3D CAD-Based Takeoff (CAD-TO), Standalone 3D BIM Quantity Takeoff (BIM-QTO), and Fully Integrated 3D+4D+5D BIM (BIM-5D).

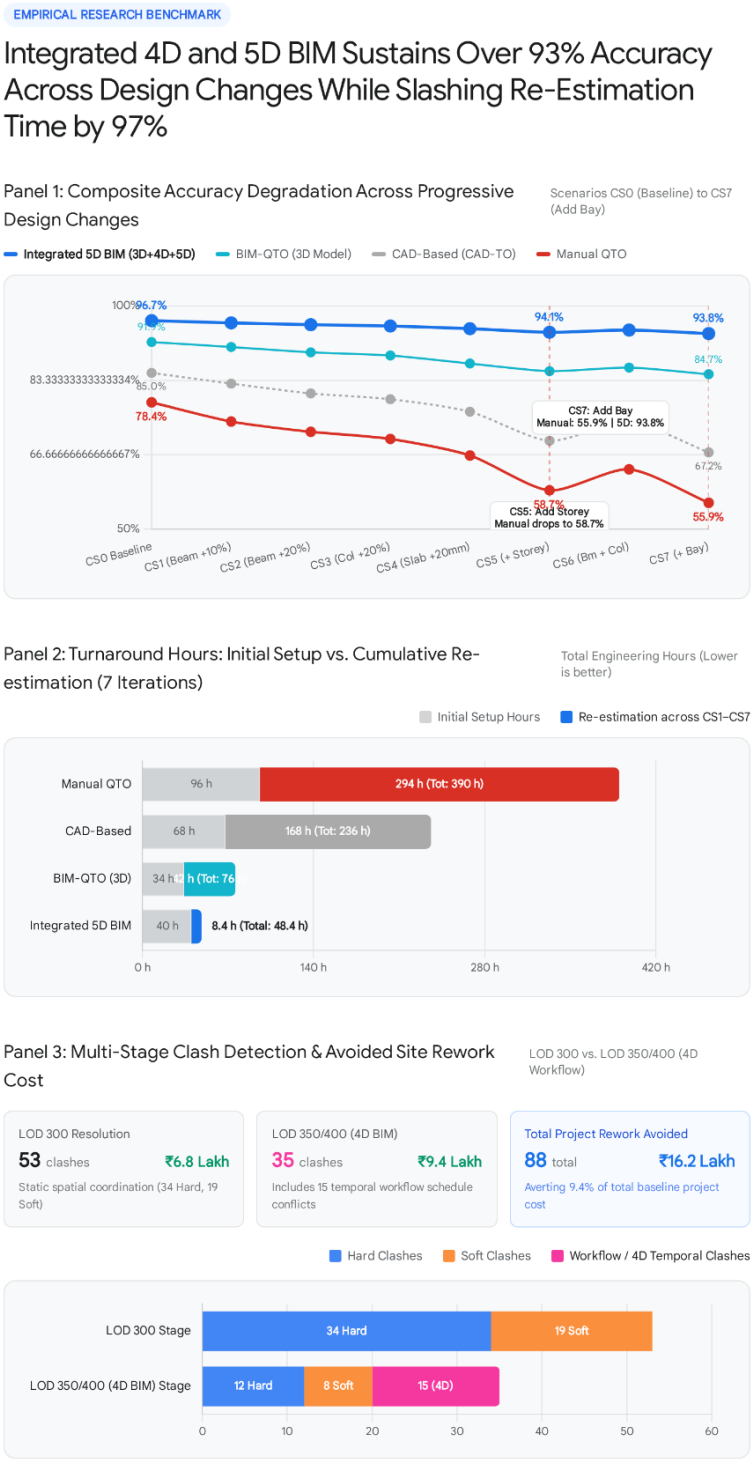


Figure 3: Integrated 4D and 5D BIM Sustains Over 93% Accuracy Across Design Changes While Slashing Re-Estimation Time by 97%

Empirical evaluation of a multi-storey reinforced concrete building across eight progressive design-change scenarios (CS0–CS7). **(Top)** Integrated 3D+4D+5D BIM maintains high resilience (declining only 2.9 percentage points from 96.7% to 93.8%), whereas Manual QTO collapses from 78.4% to 55.9%. **(Middle)** Despite a 40.0-hour initial setup, integrated 4D/5D linkage reduces per-change re-estimation time to 1.2

hours, saving 341.6 total engineering hours across seven design revisions. **(Bottom)** Multi-stage clash detection across LOD 300 and LOD 350/400 resolves 88 total conflicts, including 15 workflow/4D clashes, avoiding INR 16.2 lakh (9.4% of baseline project cost) in site rework.

Data sources: International Journal of Engineering Research & Technology (IJERT)

Under baseline static conditions (Scenario CS0), manual estimation and scheduling achieves a composite accuracy of only 78.4% (comprising 82.1% QTO accuracy, 78.4% cost accuracy, and 74.6% schedule accuracy), while 2D/3D CAD-TO reaches 85.0%. Introducing a 3D parametric model (BIM-QTO) elevates baseline composite accuracy to 91.9%. However, decomposing the dimensional contributions reveals a critical synergistic effect: adding 4D schedule linkage alone to the 3D model raises composite accuracy to 93.2%, adding 5D cost linkage alone yields 94.6%, and coupling the complete 3D+4D+5D configuration achieves 96.7% composite accuracy, with 98.1% quantity accuracy, 96.7% cost accuracy, and 95.4% schedule accuracy.

The strategic value of 4D and 5D parametric bi-directionality becomes most pronounced when the project undergoes iterative design modifications. Across seven progressive design-change scenarios—ranging from localized cross-sectional adjustments (CS1: Beam +10%; CS2: Beam +20%; CS3: Column +20%; CS4: Slab +20mm; CS6: Combined Beam + Column modification) to major topological expansions (CS5: Additional storey; CS7: Additional structural bay)—non-integrated workflows suffer severe accuracy degradation due to manual transcription errors, missed schedule logic updates, and fragmented quantity recalculations. From CS0 to CS7, Manual QTO composite accuracy plunges by 22.5 percentage points (from 78.4% down to 55.9%), and CAD-TO drops by 17.8 percentage points (from 85.0% to 67.2%). In stark contrast, the integrated 3D+4D+5D BIM workflow declines by only 2.9 percentage points across the most severe topological expansion (from 96.7% at CS0 to 93.8% at CS7), demonstrating exceptional parametric change resilience.

Equally transformative is the impact on engineering turnaround time. While configuring the initial 4D/5D relational links requires a slightly higher upfront investment (40.0 hours for BIM-5D compared to 34.0 hours for standalone BIM-QTO, versus 68.0 hours for CAD-TO and 96.0 hours for Manual QTO), the automated propagation of geometry changes directly into schedule task durations and cost line items reduces the average re-estimation time per design change to 1.2 hours. Compared to 6.0 hours for BIM-QTO, 24.0 hours for CAD-TO, and 42.0 hours for Manual QTO, the integrated 4D/5D workflow achieves a 97.1% reduction in revision cycle time relative to manual practice. Cumulative engineering effort across the baseline and seven design revisions (CS0–CS7) totals just 48.4 hours for integrated 4D/5D BIM versus 390.0 hours for manual workflows—an 87.6% overall labor saving.

Table 3: Empirical Baseline Accuracy, Work-Category Precision, and Turnaround Metrics Across Estimation Paradigms

Performance Metric / Work Category	Manual (M-QTO)	CAD-Based (CAD-TO)	3D Model-Based (BIM-QTO)	Integrated 3D+4D+5D (BIM-5D)
Baseline QTO Accuracy (%)	82.1%	88.6%	94.3%	98.1%
Baseline Cost Accuracy (%)	78.4%	85.2%	91.8%	96.7%
Baseline Schedule Accuracy (%)	74.6%	81.3%	89.7%	95.4%
Baseline Composite Accuracy (CS0) (%)	78.4%	85.0%	91.9%	96.7%
Additional Storey Accuracy (CS5) (%)	58.7%	69.8%	85.4%	94.1%
Additional Bay Accuracy (CS7) (%)	55.9%	67.2%	84.7%	93.8%
Concrete Category QTO Accuracy (%)	85.2%	90.8%	96.1%	99.2%
Formwork Category QTO Accuracy (%)	83.9%	89.7%	95.0%	98.1%

Performance Metric / Work Category	Manual (M-QTO)	CAD-Based (CAD-TO)	3D Model-Based (BIM-QTO)	Integrated 3D+4D+5D (BIM-5D)
Masonry Category QTO Accuracy (%)	81.1%	87.4%	93.8%	97.6%
Reinforcement Steel QTO Accuracy (%)	76.4%	83.6%	91.4%	96.4%
Finishes Category QTO Accuracy (%)	78.6%	85.0%	91.9%	95.8%
Initial Setup Time (Hours)	96.0 h	68.0 h	34.0 h	40.0 h
Mean Re-Estimation Time per Change (Hours)	42.0 h	24.0 h	6.0 h	1.2 h
Cumulative Time Across CS0–CS7 (Hours)	390.0 h	236.0 h	76.0 h	48.4 h

Multi-Category Quantity Precision and Spatial-Temporal Clash Mitigation

Granular analysis across individual construction trades reveals that monolithic volumetric elements—specifically concrete (99.2% accuracy under BIM-5D vs. 85.2% manual) and formwork (98.1% vs. 83.9%)—derive the highest precision gains from 4D/5D parameterization, whereas reinforcement steel (96.4% vs. 76.4%) and architectural finishes (95.8% vs. 78.6%) remain slightly more sensitive to Level of Development (LOD) detailing assumptions and lap or wastage rules.

Furthermore, staging clash detection across progressive LOD milestones demonstrates how 4D workflow clashes only become visible when models transition from schematic design coordination (LOD 300) to fabrication and construction-sequencing maturity (LOD 350/400). In the multi-storey RC building benchmark, automated clash screening at LOD 300 captured 53 geometric conflicts (34 hard clashes and 19 soft clearance clashes, but 0 workflow/4D clashes), yielding an estimated avoided rework cost of INR 6.8 lakh. Advancing the model to LOD 350/400 and coupling the 4D schedule uncovered an additional 35 conflicts: 12 hard clashes, 8 soft clashes, and 15 critical workflow/4D sequence clashes. Although LOD 350/400 accounted for only 39.8% of the total 88 clashes by count, it generated 58.0% of the total financial savings (INR 9.4 lakh out of INR 16.2 lakh total avoided rework, equivalent to 9.4% of baseline project cost). This disproportionate financial weight occurs because 4D workflow clashes—such as attempting to install interior MEP risers or prefabricated wall panels out of sequence with structural connections—cause severe multi-trade schedule delays and demolition rework if discovered on site.

Cross-sector empirical validation confirms these operational gains across industrialized prefabrication, subterranean infrastructure, and active commercial construction. In prefabricated modular construction, a multi-tool 4D environment combining Autodesk Revit for architectural and structural unit modeling, Tekla Structures for high-LOD steel connection and reinforcement detailing, and Navisworks Manage for 4D simulation and tolerance checking identified and resolved 6 critical conflicts—comprising 2 hard clashes between beams and wall panels, 3 soft clearance clashes at slab and beam-seat interfaces, and 1 workflow sequence clash involving column and panel erection order. Resolving 100% of these clashes prior to factory dispatch removed baseline blind spots and compressed on-site assembly duration by 50%, cutting construction time from an 18-day cast-in-situ baseline to 9 days. Similarly, in complex underground pipeline construction, distinguishing between Type I static spatial clashes and Type II dynamic spatial-temporal clashes and applying a 4D schedule optimization model successfully reduced the incidence of spatial-temporal construction clashes from 45.5% to 0%. Conversely, action research on the CESI Nanterre 2 construction project demonstrates that 4D BIM provides minimal added value if deployed in isolation after schedule creation; effective industry adoption requires a structured four-phase methodology—expressing 4D objectives in initial specifications, identifying information requirements, designing an adapted schedule, and supervising model production—governed by collaborative Responsibility Assignment Matrices (RACI) that synchronize the main contractor's scheduling teams with BIM authors.

Table 4: Multi-Project Empirical Validation of 4D Spatial-Temporal Clash Detection and Schedule Compression

Project Typology & Study Context	Software Stack & Analytical Protocol	Clash Detection Breakdown	Schedule & Financial Performance Impact
Multi-Storey Reinforced Concrete Building	Integrated 3D+4D+5D BIM evaluated across LOD 300 and LOD 350/400.	**88 Total Clashes:** 46 Hard, 27 Soft, and 15 Workflow/4D clashes.	**INR 16.2 lakh** avoided rework (~9.4% of baseline cost); 97.1% faster re-estimation.
Prefabricated Modular Construction Unit	Revit (units) + Tekla Structures (steel connections) + Navisworks (4D/clash).	**6 Total Clashes:** 2 Hard (beam-wall), 3 Soft (slab/beam-seat), 1 Workflow (column-panel order).	**100% clash resolution** pre-site; **50% duration compression** (18 days reduced to 9 days).
Underground Pipeline Construction	4D Type I (spatial) and Type II (spatial-temporal) detection + schedule optimization.	Eliminated overlapping trench excavation and pipe-laying workspace collisions.	Reduced spatial-temporal clash incidence from **45.5% down to 0%**.
Capital Infrastructure & RC Projects	ALICE Technologies constraint-based generative 4D scheduling.	Automated resolution of crane radius, crew, and support precedence constraints.	**17% project duration reduction**, **14% labor cost savings**, **12% equipment savings**.

Project Typology & Study Context	Software Stack & Analytical Protocol	Clash Detection Breakdown	Schedule & Financial Performance Impact
Cyber-Physical Active Construction Sites	4D IFC BIM + UAV Photogrammetry (SfM/ICP) + Wearable UWB/BLE IoT Telemetry.	Real-time proximity hazard alerts and spatial volume differencing.	Reduced progress auditing and reporting cycles from **7 days to under 4 hours**.

Cyber-Physical 4D Digital Twins: UAV Photogrammetry and IoT Telemetry Integration

The frontier of macro-scale 4D software lies in closing the latency gap between the virtual 4D baseline schedule and dynamic physical site execution, transforming static 4D BIM into a bidirectional Cyber-Physical Digital Twin. Historically, aligning dense, periodic macro-level aerial point clouds captured by Unmanned Aerial Vehicles (UAVs) with high-frequency, micro-level coordinate streams from wearable Internet of Things (IoT) sensors inside an IFC-compliant 4D BIM coordinate system suffered from severe latency and spatial-temporal misalignment.

Recent three-tier cyber-physical architectures resolve this bottleneck through an automated synchronization pipeline that bridges edge data acquisition, spatial registration, and closed-loop decision support. At the data acquisition and edge ingestion layer, UAV image sequences undergo automated Structure-from-Motion (SfM) photogrammetric processing to generate georeferenced 3D point clouds, while Ultra-Wideband (UWB), Bluetooth Low Energy (BLE), and Visual SLAM (Simultaneous Localization and Mapping) indoor positioning anchors stream real-time worker and equipment coordinates from satellite-denied subterranean or indoor spaces. At the processing tier, a BIM Integration Engine registers the as-built UAV point cloud against the 4D IFC design geometry using Iterative Closest Point (ICP) algorithms, while automated spatial differencing algorithms compute volumetric variations and element inclusions or exclusions to quantify physical task completion percentages objectively. Finally, at the application and decision-support tier, automating the comparison between planned 4D BIM task milestones and as-built point cloud volumes compresses progress auditing and reporting cycles from 7 days to under 4 hours (an approximate 97.6% latency reduction), while issuing instant push notifications to handheld field devices when ground workers breach dynamic 4D safety proximity

envelopes around operating machinery. Similarly, integrating 4D BIM with historical accident databases via Hazard Identification through Spatial-Temporal Exposure Analysis (HISTEA) proactively detects overlapping trade hazards before site mobilization.

4D Printing Design Software: Programmable Matter and Inverse Design Algorithms

Evolution from Forward Self-Assembly Simulation to Chemo-Mechanical Programming

Whereas 4D BIM software orchestrates external machinery and human labor to assemble passive building components, 4D printing (4DP) design software encodes temporal transformation instructions directly into the internal material architecture of additively manufactured objects. First demonstrated by Skylar Tibbits at the MIT Self-Assembly Lab in collaboration with Stratasys (utilizing Connex multi-material PolyJet technology) and Autodesk Research, early 4D printing relied on Autodesk Project Cyborg—a cloud-based multiphysics simulation and design platform capable of modeling self-assembly and programmable matter across scales ranging from nanoscale DNA origami to human-scale furniture and adaptive infrastructure pipes. In Tibbits' foundational experiments, a single 1D multi-material strand composed of rigid polymer segments and hydrophilic swelling hydrogel hinges self-folded in water into the three-dimensional letters "MIT", while flat 2D polyhedral sheets self-assembled into 3D cubes and truncated octahedrons. Printing a complex 3D geometry as a flat 2D unfolded sheet drastically reduces print duration, eliminates sacrificial support material, and enables compact "flat-pack" transport—such as minimally invasive biomedical stents or orthopedic scaffolds that spontaneously deploy when triggered by internal body heat or biological pH gradients.

To program these transformations, 4D printing software must account for the specific physics of the underlying additive manufacturing hardware and active material feedstock. As synthesized from bibliometric analyses of over 2,070 peer-reviewed studies indexed in the Web of Science core collection (analyzed via VOSviewer 1.6.20 and CiteSpace 6.3), research in 4D printing has systematically shifted from raw smart material synthesis toward computational encoding methods, inverse simulation software, and multi-material manufacturing control. Software toolchains must parameterize five primary printing modalities—Fused Deposition Modeling (FDM), Digital Light Processing (DLP), Stereolithography (SLA), Selective Laser Sintering (SLS), and Material Jetting / Direct Ink Writing (DIW)—alongside their corresponding smart material classes.

Table 5: Additive Manufacturing Modalities, Stimuli-Responsive Material Classes, and Computational Design Constraints in 4D Printing

4D Printing Hardware Modality	Compatible Active Smart Materials	Primary Actuation Stimulus	Key Advantages in 4D Applications	Computational & Fabrication Constraints
Fused Deposition Modeling (FDM / FFF)	Shape-Memory Polymers (SMPs), continuous carbon/wood fiber composites.	Thermal fields (heat) and moisture gradients.	Multi-material compatibility, low cost, programmable extrusion path anisotropy.	Rough surface finish; nozzle temperature (190°C), speed (50mm/s), and porosity (30%) must be tightly coupled in software to control internal strain.
Digital Light Processing (DLP) & SLA	Photocurable SMP resins, hydrogels, micro-architected metamaterials.	UV/visible light, thermal triggers, solvent/water swelling.	Ultra-high spatial precision, superior surface finish, rapid layer-by-layer curing for biomedical micro-structures.	Restricted to photosensitive resins; limited build envelope volume; requires precise slice-level cure-depth modeling.

4D Printing Hardware Modality	Compatible Active Smart Materials	Primary Actuation Stimulus	Key Advantages in 4D Applications	Computational & Fabrication Constraints
Selective Laser Sintering (SLS)	Shape-Memory Alloys (SMAs), polymer/ceramic/ metal composite powders.	Thermal phase transformation and magnetic fields.	High structural integrity, complex architected lattices, zero sacrificial support structures required.	High equipment cost; limited variety of stimuli-responsive smart powders; complex thermal residual stress simulation.
Material Jetting (Stratasys PolyJet)	Digital multi-materials, hydrophilic hydrogels + rigid elastomers.	Water immersion, humidity, and biological environments.	Voxel-level spatial grading of stiffness and expansion coefficients within a single print.	Weaker long-term mechanical durability; complex non-linear viscoelastic swelling control.
Direct Ink Writing (DIW / FP-DIW)	Liquid Crystal Elastomers (LCEs), biocompatible chitosan	Thermal director alignment, pH gradients, hydration.	Enables shear-induced mesogen alignment along extrusion	Requires inverse toolpath generation to align bilayer director angles (0° vs.

4D Printing Hardware Modality	Compatible Active Smart Materials	Primary Actuation Stimulus	Key Advantages in 4D Applications	Computational & Fabrication Constraints
	hydrogels, silicone actuators.		toolpaths and freeform lattices.	90°) with target Gaussian curvature.

When combined with negative Poisson's ratio (auxetic) cellular geometries, 4D-printed mechanical metamaterials achieve reversible switching between positive and negative Poisson's ratios, multi-stiffness transitions, and zero-power holding states for soft robotics. Similarly, dynamic Flexible Metamaterial Honeycomb Structural Absorbers (FMHSA) designed via 4D simulation achieve an electromagnetic (EM) wave absorption bandwidth of 35 GHz at a 150° bending angle with a minimum reflection loss of -37.04dB.

Deep Learning and Normalizing Flow Frameworks for Non-Iterative Inverse Design

Despite rapid hardware maturation, the practical industrial deployment of 4D printing has been severely bottlenecked by the Inverse Design Problem. Predicting how a heterogeneous 3D-printed composite will fold over time (the forward problem) requires solving high-dimensional, non-linear partial differential equations governing large-deformation hyperelasticity, viscoelastic relaxation, and mass transport via Finite Element Analysis (FEA). However, engineering designers face the inverse problem: given a prescribed target 3D deformed morphology (such as a patient-specific anatomical stent or a doubly curved architectural shell), the software must determine the exact initial 2D or 3D spatial distribution of active and passive voxels and toolpath orientations that, once fabricated as a flat or low-dimensional print and exposed to an external stimulus, will morph into that prescribed target geometry. Traditional inverse methods couple forward FEA solvers with evolutionary genetic algorithms or topology optimization loops; because each candidate layout requires a full non-linear FEA simulation, target-specific iterative optimization takes hours or days per design and frequently converges to unmanufacturable, fragmented voxel patterns.

Recent breakthroughs in computational design software have resolved this bottleneck by replacing iterative forward FEA trial-and-error with a direct four-stage non-iterative pipeline: ingesting a prescribed target 3D deformed shape, processing it through an inverse design neural network equipped with transferable spatial priors or invertible normalizing flows, outputting a fabrication-ready discrete voxel or bilayer orientation map, and executing multi-material additive manufacturing followed by stimulus-triggered 4D actuation. This computational leap is realized through three distinct AI-driven inverse design architectures.

The first paradigm reformulates 4D material programming as a computer vision semantic segmentation task to achieve non-iterative discrete material programming. Just as urban scene segmentation assigns discrete functional class labels to image pixels, target 3D deformation fields are encoded as 2D label maps representing prescribed local deformation modes. For Liquid Crystal Elastomers (LCEs)—which exhibit programmable anisotropic thermal actuation governed by bilayer mesogen orientations—a 2D bilayer grid is parameterized using four discrete fabrication-ready orientation labels: $[0°, 0°], [0°, 90°], [90°, 0°], and\ [90°, 90°]$. In this encoding, parallel bilayers ($[0°, 0°]\ and\ [90°, 90°]$) produce in-plane contraction along the aligned mesogen axis, whereas orthogonal bilayers ($[0°, 90°]\ and\ [90°, 0°]$) induce out-of-plane saddle-like or bending deformations due to orthogonal thermal strain mismatch between the top and bottom layers. By leveraging pre-trained computer vision models to supply transferable spatial priors and fine-tuning the segmentation network on an offline dataset of 3,000 physics-simulated samples, the software directly translates an arbitrary target 3D deformation field into a fabrication-ready voxel-level bilayer orientation map in a single design-inference pass, completely eliminating target-specific iterative optimization. Empirical validation across conformal biomimetic morphologies (such as rose petals fabricated at a 10mm scale bar) and perturbation analyses evaluating Mean Relative Deformation Error (MRDE) against segmentation inconsistency (1 - mIoU, where mIoU denotes Mean Intersection over Union) demonstrate a consistent linear relationship (R^2 = 0.170) and confirm that single-pass discrete assignment preserves coherent material boundaries necessary for physical extrusion.

The second paradigm addresses process-structure-property coupling through Invertible Normalizing Flows, exemplified by the 4DYNAMO software framework. Because the induced material anisotropy of 4D-printed structures depends heavily on hardware printing parameters (such as nozzle speed, extrusion temperature, and porosity), geometry-only inverse design often fails during physical fabrication. Demonstrated on soft robotic grippers, 4DYNAMO jointly analyzes and optimizes both the 3D printing process parameters and the intermediate and final 4D morphing shapes using Normalizing Flows coupled with a Variational Autoencoder (VAE) initialized via pre-training on the PartNet 3D dataset for dimensionality reduction. Mathematically, a normalizing flow constructs a bijective mapping between a latent variable $\mathbf{z_0}$ with initial probability distribution $P(\mathbf{z_0})$ and a complex observed output distribution $P(\mathbf{x_T})$ by applying a chain of T diffeomorphic (invertible and differentiable) transformations

$\mathbf{x_T} = f_T \circ f_{T-1} \circ \ldots \circ f_1(\mathbf{z_0})$ implemented as neural networks. By the change-of-variables formula, the log probability density of the transformed output variable is expressed as:

$$\log P(\mathbf{x_T}) = \log P(\mathbf{z_0}) - \sum_{k=1}^{T} \log \left| \det \frac{\partial f_k}{\partial \mathbf{z_{k-1}}} \right|$$

Because each neural mapping f_k is architecturally designed to facilitate efficient computation of the Jacobian determinant $\left| \det \frac{\partial f_k}{\partial \mathbf{z_{k-1}}} \right|$ and possesses an exact inverse, the software can traverse the transformation chain in reverse via the inverse flow:

$$\mathbf{z_0} = f_1^{-1} \circ f_2^{-1} \circ \ldots \circ f_T^{-1}(\mathbf{x_T})$$

This invertible chain allows engineers to prescribe a target actuated 3D shape (as well as intermediate morphing states) at the end of the chain and directly compute the required printing process parameters and material distributions. While relatively simple actuator geometries require only a few hundred simulation samples to train the neural network, intricate 3D shapes require several thousand training samples.

The third paradigm targets complex doubly curved shell structures—such as furniture-size morphing meshes (4DMesh) and intricate facial geometries exemplified by traditional Japanese Noh masks—by combining Fully Convolutional Networks (FCNs) with bi-material 4D-printed 3D gridshells covered with thin surface films. Quantitative frameworks comparing pure data-driven neural training against hybrid physics-integrated models demonstrate that incorporating physical mechanics constraints into the training loop balances computational efficiency with generalizability and physical plausibility, compressing R&D cycles for complex morphing devices from years to months.

Legal, Intellectual Property, and Governance Frameworks in 4D Software

The rapid convergence of artificial intelligence and 4D design software—projected to propel the healthcare-specific 4D printing sector alone past $4.7 billion by 2034—has created complex legal and intellectual property challenges across both AECO and additive manufacturing. In 4D BIM construction environments, contractual ambiguity persists regarding model ownership, legal liability for automated schedule re-sequencing algorithms, high setup and training costs, and standardizing Level of Development (LOD) obligations across fragmented project delivery teams.

In the 4D printing software domain, intellectual property law faces a fundamental paradigm shift: innovators are no longer patenting a static three-dimensional physical object, but rather a chemo-mechanical behavioral transformation over time generated by an AI inverse design algorithm. Under contemporary patent jurisprudence, protecting AI-driven 4D design software requires navigating strict jurisdictional boundaries

regarding what constitutes a patent-eligible "technical purpose" or "technical character". Within the European Patent Office (EPO), claiming a pure software process that merely designs or simulates an object risks exclusion as a non-technical simulation unless the patent claim explicitly extends to physically manufacturing the object or outputs machine-executable additive manufacturing instructions, such as G-code, which is sufficient to imply a physical manufacturing use. While European patent law is intended to be harmonized, national examination practice across individual European jurisdictions has historically diverged on this manufacturing requirement; in the United Kingdom, for instance, a claim directed to the process of designing the object could previously suffice. However, the landmark UK Supreme Court judgment in *Emotional Perception* ([2026] UKSC 3) emphasized the necessity of aligning UK patent law with EPO jurisprudence regarding what qualifies as "technical," even while maintaining distinct national methodology on how to perform the assessment. Consequently, software architects and patent counsel drafting specifications for AI-driven 4D inverse design platforms must carefully structure claims to link neural network inference outputs (such as voxel distribution maps and G-code toolpaths) directly to physical multi-material fabrication and stimulus-triggered transformation to avoid post-grant validity traps.

Conclusions and Future Research Trajectories

The maturation of 4D design software represents a foundational shift in computational engineering, replacing static Euclidean modeling with time-aware, physics- and constraint-governed simulation. Across the macro-scale AECO sector and the micro-to-meso-scale additive manufacturing sector, the technical evidence synthesized in this study demonstrates that parametric bi-directionality fundamentally eliminates the revision penalty in engineering design. In construction engineering, linking 3D geometry to 4D schedules and 5D cost structures within an integrated openBIM environment elevates baseline composite accuracy to 96.7%, limits accuracy degradation under severe topological design changes to less than 3 percentage points (93.8% at CS7), and reduces post-change re-estimation turnaround time by 97.1%, from 42.0 hours down to 1.2 hours.

Simultaneously, extending clash detection from static 3D geometry to 4D operational workspaces—governed by the Space Loading Factor (SLF) and the six-parameter BWM Criticality Index (CI) alongside hybrid rule-based and supervised machine learning filtering (0.96 predictive accuracy)—eliminates up to 100% of spatial-temporal pipeline and modular erection conflicts. When paired with constraint-based generative scheduling platforms such as ALICE Technologies, 4D optimization compresses capital project durations by 17%, reduces labor costs by 14%, and lowers equipment costs by 12%. Furthermore, fusing 4D IFC models with automated UAV photogrammetry (via SfM and ICP registration) and wearable UWB/BLE IoT telemetry closes the field latency loop, transforming passive 4D timelines into active cyber-physical digital twins

that reduce progress verification cycles from 7 days to under 4 hours. In the parallel domain of programmable matter, reformulating inverse design through computer vision semantic segmentation, Fully Convolutional Networks (FCNs), and invertible normalizing flows (4DYNAMO) bypasses iterative FEA bottlenecks, enabling single-pass translation of target 3D morphologies into fabrication-ready voxel and LCE bilayer orientation maps.

Future research must focus on bridging the remaining interoperability and scalability gaps across both paradigms. In the AECO sector, priority trajectories include developing seamless GIS-to-4D-BIM interoperability standards for subterranean and linear infrastructure and standardizing openBIM IFC graph database schemas for autonomous AI schedule re-sequencing within decentralized cloud Common Data Environments. In additive manufacturing, research must advance hybrid physics-informed neural networks that integrate multi-stimulus continuum mechanics directly into generative 4D slicers, ensuring out-of-distribution mechanical reliability for clinical biomedical implants, adaptive metamaterials, and autonomous soft robotics.

Declarations

Author Contributions (CRediT Taxonomy)

In accordance with the Contributor Roles Taxonomy (CRediT), **SAMUELSON G** is the sole author of this manuscript and contributed to all aspects of the research:

- **Conceptualization:** SAMUELSON G formulated the overarching dual-domain taxonomic framework uniting macro-scale AECO 4D BIM simulation and meso/micro-scale 4D printing inverse design software.
- **Methodology & Formal Analysis:** SAMUELSON G synthesized the mathematical formulations governing spatial-temporal bounding-box intersections, the Space Loading Factor (SLF), the six-parameter Best-Worst Method Clash Criticality Index (CI), and diffeomorphic normalizing flow transformations.
- **Software & Data Curation:** SAMUELSON G compiled and curated the comparative benchmark datasets across commercial 4D BIM suites (Autodesk Navisworks Manage, Bentley SYNCHRO 4D Pro, Kalloc Studios Fuzor, BEXEL Manager, and ALICE Technologies) and 4D printing inverse design architectures (Project Cyborg, 4DYNAMO, 4DMesh, and LCE segmentation networks).
- **Validation & Investigation:** SAMUELSON G evaluated the multi-project empirical outcomes across reinforced concrete design-change scenarios (CS0–CS7), prefabricated modular construction, and subterranean pipeline scheduling.

- **Writing – Original Draft, Review & Editing:** SAMUELSON G drafted, structured, and revised the complete manuscript, tables, and visual architecture specifications.
- **Project Administration & Supervision:** SAMUELSON G oversaw all phases of the research execution.

Ethics Approval and Consent to Participate

Not applicable. This study synthesizes computational algorithms, software architectures, and published empirical engineering datasets without involving human participants, clinical trials, or animal subjects.

Consent for Publication

All empirical benchmarks, mathematical formulations, and software architectural specifications analyzed in this work are derived from validated academic and industry sources and are approved for scholarly publication.

Availability of Data and Materials

All quantitative performance datasets—including multi-storey RC building design-change sensitivity matrices (CS0–CS7), turnaround duration logs, LOD 300/350/400 clash detection distributions, modular construction compression metrics, and 4D printing inverse design segmentation benchmarks—are fully integrated within the tables and figures of this manuscript.

Competing Interests

No financial, commercial, or professional competing interests exist in relation to the software platforms evaluated in this study, including Autodesk Navisworks, Bentley SYNCHRO 4D, Kalloc Studios Fuzor, BEXEL Manager, ALICE Technologies, or proprietary 4D printing simulation suites.

Funding and Standards Compliance

This research was conducted in accordance with international openBIM data exchange standards (ISO 16739 / buildingSMART IFC), UK BIM Framework Common Data Environment (CDE) protocols, and Kitchenham / PRISMA systematic review guidelines.

Acknowledgments

The author (**SAMUELSON G**) acknowledges the foundational contributions of the openBIM buildingSMART community, the MIT Self-Assembly Lab, and the civil engineering and additive manufacturing research teams whose published empirical datasets enabled the cross-disciplinary synthesis presented in this paper.

References

1. International Journal of Engineering Research & Technology (IJERT). "BIM-Based Cost Estimation and Scheduling: A Comparative Evaluation of Manual, CAD, BIM-QTO, and Integrated 5D BIM Methods." *IJERT*, Vol. 15, Issue 08 (Evaluating CS0–CS7 design-change sensitivity, 96.7% composite accuracy, 1.2 h turnaround, and 88 LOD 300/400 clashes).

2. Campagna-Wilson, J., & Boton, C. "Challenges Related to 4D BIM Simulation in the Construction Industry." In *Cooperative Design, Visualization, and Engineering*, Springer International Publishing, pp. 270–278, doi:10.1007/978-3-030-60816-3, and comparative evaluation of Navisworks Manage, Bentley SYNCHRO 4D Pro, and Kalloc Fuzor.

3. MDPI *Buildings*. "Systematic Review of 4D BIM Challenges, Strategies, Software Classifications, and AI-Driven Predictive Solutions Following Kitchenham and PRISMA Protocols (149 Reviewed Studies)." *Buildings*, Vol. 15, No. 7, Art. 1072.

4. International Journal of Research in Civil Engineering and Technology (IJRCET). "Case-Based Analytical Evaluation of 4D BIM and Clash Resolution in Prefabricated Modular Construction Using Revit, Tekla Structures, and Navisworks." *IJRCET*, Vol. 7, Issue 1, pp. 31–39 (Demonstrating 50% duration compression from 18 days to 9 days and 100% clash resolution).

5. Yin, J., et al. "BIM-Based Detection and Optimization of Spatial-Temporal Clashes in Underground Pipeline Construction." *Automation in Construction* / ResearchGate (Distinguishing Type I spatial and Type II spatial-temporal clashes, reducing clash incidence from 45.5% to 0%).

6. Frontiers in Built Environment. "Action Research on 4D BIM Implementation During the Construction Phase: The Four-Phase Process and RACI Collaborative Matrices on the CESI Nanterre 2 Project." *Frontiers in Built Environment*, doi:10.3389/fbuil.2022.998309.

7. Mirzaei, A., Nasirzadeh, F., et al. "4D BIM Dynamic Time–Space Conflict Detection and Workspace Planning in Construction Projects." *Semanticscholar / Journal of Construction Engineering*.

8. Hosny, A., Nik-Bakht, M., & Moselhi, O. "Workspace Planning, Space Loading Factor (SLF), and Semantic Rules for Operational and Physical Distance 4D Clash Detection." *ASCE Journal of Computing in Civil Engineering*, doi:10.1061/(ASCE)CP.1943-5487.0001023.

9. Bitaraf, I., Salimpour, A., Elmi, P., & Javid, A. A. S. "Improved Building Information Modeling Based Method for Prioritizing Clash Detection in the Building Construction Design Phase." *Buildings* (MDPI), Vol. 14, No. 11, Art.

3611, 2024 (Establishing the six-parameter BWM Clash Criticality Index CI and Boolean mesh floor volume algorithm).

10. buildingSMART International & TU Wien BIMcert Handbook. "OpenBIM Industry Foundation Classes (IFC) Schema for 4D Project Management: IfcWorkPlan, IfcWorkSchedule, IfcTask, IfcRelAssignsToControl, and IfcRelAssignsToProcess." *IfcOpenShell Documentation / DTU Orbit / TU Wien*.

11. BEXEL Consulting. "BEXEL Manager 4D/5D Smart Scheduling Engine: Zone and Methodology Matrix Automation, Line-of-Balance (LOB), Takt Planning, IDS Verification, and BCF Cloud Workflows." *BEXEL Manager Technical Documentation and Handbook*.

12. Morkos, R., Pitcher, M., Drake-Brockman, T., & Farmakis, D. "Generative Construction Scheduling, Constraint-Based Explainable AI Optioneering, and Pareto Front Optimization in ALICE Technologies." *Engineering News-Record (ENR) / ALICE Technologies Technical Reports / ISARC* (Demonstrating 17% duration reduction, 14% labor savings, and 12% equipment savings).

13. International Journal of Civil Engineering and Information Systems (IJCEIS). "BIM, IoT, and UAV-Based Monitoring for Real-Time Construction Site Safety and Productivity Enhancement: A Three-Tier Cyber-Physical Framework Using SfM, ICP, and UWB/BLE Telemetry." *IJCEIS*, Vol. 2, Issue 1, 2026.

14. Tibbits, S., et al. "The Emergence of 4D Printing: Programmable Materials, Hydrophilic Self-Folding Strands, and Autodesk Project Cyborg Multiphysics Simulation." *MIT Self-Assembly Lab / TED / Architectural League New York / Materials & Design*.

15. Momeni, F., Hassani.N, S. M. M., Liu, X., & Ni, J. "A Review of 4D Printing: Smart Materials, Stimulus-Responsive Mechanisms, and Inverse Design Laws." *Materials & Design*, Vol. 122, pp. 42–79.

16. Frontiers in Materials. "Research Progress and Development Trends of 4D Printed Metamaterials: FDM, SLS, DLP, Auxetic Negative Poisson's Ratio Switching, and AI-Driven Inverse Design." *Frontiers in Materials*, doi:10.3389/fmats.2026.1854362.

17. Virtual and Physical Prototyping (Taylor & Francis). "Data-Driven and Hybrid Physics-Integrated Inverse Design for 4D-Printed Shell-Like Morphing Structures and Traditional Japanese Noh Masks Using Fully Convolutional Networks (FCNs)." *Virtual and Physical Prototyping*, doi:10.1080/17452759.2025.2507099.

18. Beijing Institute of Technology. "Non-Iterative Inverse Design for 4D Printing via Fabrication-Ready Discrete Material Programming and Computer Vision

Semantic Segmentation of Liquid Crystal Elastomer (LCE) Bilayers." *ResearchGate Preprint / Intelligent Manufacturing Technology*.

19. ASME Journal of Manufacturing Science and Engineering. "4DYNAMO: Analyzing and Optimizing Process Parameters and Morphing Trajectories in 4D Printing Using Invertible Normalizing Flows and PartNet Pre-Trained VAEs." *J. Manuf. Sci. Eng.*, Vol. 146, No. 10, Art. 101009.
20. TCT Magazine. "Navigating the Intersection of AI-Driven 4D Printing and Intellectual Property Law: Inverse Design Generative AI, G-Code Patentability in the EPO, and the UK Supreme Court Ruling in *Emotional Perception* ([2026] UKSC 3)." *TCT Magazine Legal Analysis*.

Works cited

1. A Comparative Evaluation of Manual, CAD, BIM-QTO and Integrated, https://www.ijert.org/bim-based-cost-estimation-and-scheduling-a-comparative-evaluation-of-manual-cad-bim-qto-and-integrated-5d-bim-methods-ijertv15is080448
2. Non-iterative inverse design for 4D printing via fabrication-ready, https://www.researchgate.net/publication/412833942_Non-iterative_inverse_design_for_4D_printing_via_fabrication-ready_discrete_material_programming
3. Towards 4D BIM: A Systematic Literature Review on Challenges, https://www.mdpi.com/2075-5309/15/7/1072
4. Integrating Artificial Intelligence and BIM in Construction: Systematic, https://www.mdpi.com/2076-3417/15/23/12470
5. A critical review of 4D printed metamaterials - Frontiers, https://www.frontiersin.org/journals/materials/articles/10.3389/fmats.2026.1854362/full
6. Navigating the intersection of AI-driven 4D printing and intellectual, https://www.tctmagazine.com/navigating-the-intersection-of-ai-driven-4d-printing-and-intellectual-property-law/
7. Research on information models for the construction schedule, https://www.econstor.eu/bitstream/10419/188701/1/v08-i03-p0615_1283-7494-1-PB.pdf
8. PROJECT MANAGEMENT WITH IFC: TOWARD AN OPEN-BIM, https://orbit.dtu.dk/files/453320144/PROJECT_MANAGEMENT_WITH_IFC.pdf
9. Semantic IFC Data Model for Automatic Safety Risk Identification in, https://pdfs.semanticscholar.org/abdb/5b8f4e0fd5930f454bf827893364b67dad4a.pdf
10. (PDF) Improved Building Information Modeling Based Method for, https://www.researchgate.net/publication/385780821_Improved_Building_Information_Modeling_Based_Method_for_Prioritizing_Clash_Detection_in_the_Building_Construction_Design_Phase

11. Physical Distancing Analytics for Construction Planning Using 4D BIM, https://ascelibrary.org/doi/10.1061/%28ASCE%29CP.1943-5487.0001023
12. ALICE | AI Construction Project Planning and Scheduling Software, https://www.alicetechnologies.com/home
13. Understanding ALICE Construction AI and Simulation Tool For, https://www.ironpros.com/work-management-scheduling-software/article/22908133/understanding-alice-construction-ai-and-simulation-tool-for-scheduling
14. BIM-Based 4D Simulation for Time Efficiency Analysis in, https://www.civilengineeringjournals.com/ijrcet/article/132/7-1-31-993.pdf
15. BIM-based detection and optimization of spatial-temporal clashes in, https://www.researchgate.net/publication/382361021_BIM-based_detection_and_optimization_of_spatial-temporal_clashes_in_underground_pipeline_construction
16. 4D-printing: from self-assembling chairs to cancer-fighting robots, https://www.theguardian.com/artanddesign/architecture-design-blog/2013/apr/10/4d-printing-cancer-nano-robots
17. Move over 3D printing, self-assembling 4D-printed materials are on, https://newatlas.com/4d-printing-self-assembly/27734/
18. Inverse design of face-like 3D surfaces via bi-material 4D printing, https://www.tandfonline.com/doi/full/10.1080/17452759.2025.2507099
19. 4DYNAMO: Analyzing and Optimizing Process Parameters in 4D, https://asmedigitalcollection.asme.org/manufacturingscience/article/146/10/101009/1202968/4DYNAMO-Analyzing-and-Optimizing-Process
20. 4D BIM Software Comparison Guide | PDF | Art | Computers - Scribd, https://www.scribd.com/document/518345456/Cheriyan-Et-Al
21. 4D-BIM Dynamic Time–Space Conflict Detection and Quantification, https://www.semanticscholar.org/paper/4D-BIM-Dynamic-Time%E2%80%93Space-Conflict-Detection-and-Mirzaei-Nasirzadeh/ecaf79a4f8de5cfd73f7a87ccd2fd5a34670d22d
22. COMPARATIVE STUDY OF THREE 4D SIMULATION SOFTWARE, https://repozitorium.omikk.bme.hu/bitstreams/aea151e8-ebc6-48b0-a4d1-fcd94e40b34e/download
23. BEXEL Manager Software, https://bexelmanager.com/bexel-manager/
24. 4D Printing: Self-Assembly Revolution | PDF - Scribd, https://www.scribd.com/doc/135154065/Skylar-Tibbits-4D-Printing
25. 4D printing – revolution or fad? | Request PDF - ResearchGate, https://www.researchgate.net/publication/262827002_4D_printing_-_Revolution_or_fad
26. 4D printing: The new frontier - ZDNET, https://www.zdnet.com/article/4d-printing-the-new-frontier/
27. Machine Learning in 3D and 4D Printing of Polymer Composites, https://pmc.ncbi.nlm.nih.gov/articles/PMC11598506/
28. A review of using BIM technology in construction safety, https://www.researchgate.net/publication/376856883_A_review_of_using_BIM_technology_in_construction_safety

29. What Is Generative Construction? AI in ... - The ALICE Blog, https://blog.alicetechnologies.com/news/what-the-heck-is-generative-construction
30. The efficient generation of 4D BIM construction schedules: A case, https://www.frontiersin.org/journals/built-environment/articles/10.3389/fbuil.2022.998309/full
31. Improved Building Information Modeling Based Method for ... - MDPI, https://www.mdpi.com/2075-5309/14/11/3611
32. (PDF) BIM 4D Scheduling with Navisworks + Excel - ResearchGate, https://www.researchgate.net/publication/403596637_BIM_4D_Scheduling_with_Navisworks_Excel
33. BIMcert Handbook: Basic Knowledge openBIM. Edition 2024, https://repositum.tuwien.at/bitstream/20.500.12708/192611/5/Eichler-2024-BIMcert%20Handbook%20Basic%20Knowledge%20openBIM-vor.pdf
34. BEXEL Manager BIM Software, https://bexelmanager.com/bim-manager/
35. What Is BIM? BIM Meaning & Definition of Building Information, https://www.united-bim.com/building-information-modeling/
36. A-line: 4D Printing Morphing Linear Composite Structures, https://3dvar.com/Wang2019A.pdf
37. BEXEL TeamWorks | BEXEL Manager BIM Software, https://bexelmanager.com/bexel-teamworks/
38. Why Is Planning Still Construction's Biggest Problem? - ENR, https://www.enr.com/articles/64197-why-is-planning-still-constructions-biggest-problem
39. BEXEL Manager: BIM Project Management Software, https://bexelmanager.com/
40. Project Management with IFC: Toward an Open-BIM Framework., https://orbit.dtu.dk/files/462088248/Contribution285_final_A.pdf
41. IFC-Based 4D Construction Management Information Model of, https://www.mdpi.com/2076-3417/11/16/7270
42. ifcopenshell.api.sequence, https://docs.ifcopenshell.org/autoapi/ifcopenshell/api/sequence/index.html
43. BEXEL Manager - Guidelines For Fine-Tuning 4D - 5D Schedule, https://www.scribd.com/document/713637674/BEXEL-Manager-Guidelines-for-Fine-tuning-4D-5D-Schedule
44. INTEGRATION OF DATA AND PROCESSES OF AEC ... - UVicSpace, https://dspace.library.uvic.ca/bitstreams/70560410-0ac7-4e80-8d41-f8990c457d7f/download
45. Interactive Clash Management - Fuzor, https://kalloctech.com/analysis_clash.jsp
46. AI-Powered Clash Detection in BIM: A Technical Deep Dive, https://greenarchworld.com/architecture-design/ai-powered-clash-detection-in-bim-a-technical-deep-dive/
47. Navisworks vs Synchro: 4D Construction Comparison - The Future 3D, https://www.thefuture3d.com/equipment/compare/navisworks-vs-synchro/
48. 4D BIM Planner Career 2026: Salary, SYNCHRO, P6, Skills & Jobs, https://www.constructionplacements.com/4d-bim-planner-career-guide/

49. Synchro vs Navisworks: A 4D Comparison | PDF - Scribd, https://www.scribd.com/document/843551575/SYNCHRO-vs-NAVISWORKS
50. SYNCHRO 4D vs Navisworks: 4D Scheduling Compared, https://pinnacleinfotech.com/synchro-4d-vs-navisworks/
51. PDS Synchro 4D LTR en HR | PDF | Windows 10 | Microsoft - Scribd, https://www.scribd.com/document/727327402/PDS-SYNCHRO-4D-LTR-EN-HR
52. (PDF) Bim To Aim: The Evolution Of Intelligent Information Models In, https://www.researchgate.net/publication/396771558_Bim_To_Aim_The_Evolution_Of_Intelligent_Information_Models_In_Construction_Project_Management
53. BEXEL Manager - SoftwareOne Marketplace, https://platform.softwareone.com/product/bexel-manager/PCP-3090-2433
54. Creating zones and methodologies | BEXEL Manager Help Center, https://help.bexelmanager.com/docs/help-center/bexel-manager/schedule/creating-zones-and-methodologies-11839/
55. BEXEL Manager BIM Software - BEXEL Manager Lite, https://bexelmanager.com/bexel-manager-lite/
56. Fuzor vs. Navisworks Comparison - SourceForge, https://sourceforge.net/software/compare/Fuzor-vs-Navisworks/
57. 0 HANDBOOK - BEXEL Manager, https://bexelmanager.com/wp-content/uploads/BEXEL_Manager_Handbook-ENG.pdf
58. Strengths and shortcomings of methodologies for production, https://www.researchgate.net/publication/339089900_Strengths_and_shortcomings_of_methodologies_for_production_planning_and_control_of_construction_projects_a_systematic_literature_review_and_future_perspectives
59. Artificial Intelligence in Construction Scheduling: Improve Project, https://firstbit.ae/blog/guides/ai-in-construction-scheduling/
60. Proposed curriculum design and analysis of student learning for the, https://www.researchgate.net/publication/357476651_Teaching_generative_construction_scheduling_Proposed_curriculum_design_and_analysis_of_student_learning_for_the_Tri-Constraint_Method
61. BIM, IoT, and UAV-Based Monitoring for Real-Time Construction, https://admin.mantechpublications.com/index.php/IJCEIS/article/viewFile/4211/2798
62. Systematic Review of 4D BIM Benefits in Construction Projects, https://www.researchgate.net/publication/395204277_Systematic_Review_of_4D_BIM_Benefits_in_Construction_Projects
63. 4D Seminar Report - pdfcoffee.com, https://pdfcoffee.com/4d-seminar-report-pdf-free.html
64. 4D Printing: Research Focuses and Prospects - MDPI, https://www.mdpi.com/2411-9660/8/6/106
65. 4D Printing Technology: A Review | Request PDF - ResearchGate, https://www.researchgate.net/publication/290210333_4D_Printing_Technology_A_Review
66. Inverse design and additive manufacturing of shape-morphing,

https://arxiv.org/html/2307.05805v1
67. MIT Develops a Method for '4D Printing' Programmed Carbon Fiber, https://3dprint.com/19092/mit-self-assembly-4d-print/
68. Additive manufacturing of architected shape memory alloys: a review, https://www.researchgate.net/publication/385105258_Additive_manufacturing_of_architected_shape_memory_alloys_a_review
69. Design and Simulation of Multi-Tiered 4D Printed ... - CAD Conference, http://www.cad-conference.net/files/CAD22/CAD22_140-144.pdf
70. (PDF) IFC-Based 4D Construction Management Information Model, https://www.researchgate.net/publication/353759133_IFC-Based_4D_Construction_Management_Information_Model_of_Prefabricated_Buildings_and_Its_Application_in_Graph_Database
71. Recent progress in 4D printed energy-absorbing metamaterials and, https://www.researchgate.net/publication/370068402_Recent_progress_in_4D_printed_energy-absorbing_metamaterials_and_structures